

Search-Tree Topology Collapse under Rule Interventions: Severe Testing of Symbolic World Models in Game AI

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Abstract

Model-based artificial intelligence relies on world models to forecast state transitions and plan action sequences. While model learning traditionally optimizes one-step predictive likelihood or passive transition accuracy (A_{pred}), high accuracy on observation traces does not guarantee plan validity during active goal-directed search—a phenomenon known in continuous domains as *objective mismatch*. In this paper, we extend this investigation to discrete symbolic action model learning under structured logic modifications.

We formulate a formal severe testing framework anchored on state-indexed search trees, defining the *Edge Symmetric Distance* (d_{Δ}) and topological cut-weight $\omega(p^*)$. We prove Theorem 1 (Non-Sufficiency of Passive Accuracy): constructing a 100-state discrete grid domain M where two learned models $\hat{M}_A$ and $\hat{M}_B$ achieve identical passive accuracy $A_{\text{pred}} = 98.0\%$, yet omitting a leaf precondition in $\hat{M}_A$ yields $d_{\Delta} = 1$ and successful plan execution ($R_{\text{play}} = 0$), whereas omitting a bottleneck precondition in $\hat{M}_B$ causes complete search-tree topology collapse ($d_{\Delta} = 99$) and infinite play regret ($R_{\text{play}} = \infty$).

To empirically validate our framework on real game engines, we compile authoritative NodeJS PuzzleScript games (Sokoban, It Is Pitch Black, Graded Sir) into PDDL action schemas and evaluate symbolic learners (FastLAS, WorldCoder, SLAF, FAMA, LOCM2). Through a trajectory bisimulation validation protocol ($100.0\% \pm 0.0\%$ match across 5,000 test traces), we demonstrate that passive accuracy fails to detect bottleneck rule omissions, whereas Answer Set Programming (FastLAS) preserves search-tree topology ($d_{\Delta} = 13.57 \pm 2.29$) due to non-monotonic logical constraint retention.

Keywords: Symbolic World Models, Action Model Learning, Search-Tree Topology, Rule Interventions, Objective Mismatch, Severe Testing, Game AI

1. Introduction

Building faithful dynamics models of environments is a foundational objective of model-based planning and reinforcement learning [1, 2]. When an agent possesses a correct world model, it can simulate future state transitions, evaluate counterfactual trajectories, and synthesize plan sequences using heuristic search algorithms such as A^* or Monte Carlo Tree Search (MCTS).

Traditionally, symbolic action model learning algorithms—such as LOCM2 [3], FAMA [1], FastLAS [4], ARMS [5], and SLAF [6, 7]—evaluate performance by measuring Schema Accuracy (SA) or Passive Transition Accuracy (A_{pred}) over historical execution traces. However, evaluating a world model solely on passive prediction introduces a severe vulnerability: global transition accuracy treats all state transitions uniformly, whereas search-tree planning is non-uniformly sensitive to critical bottleneck preconditions.

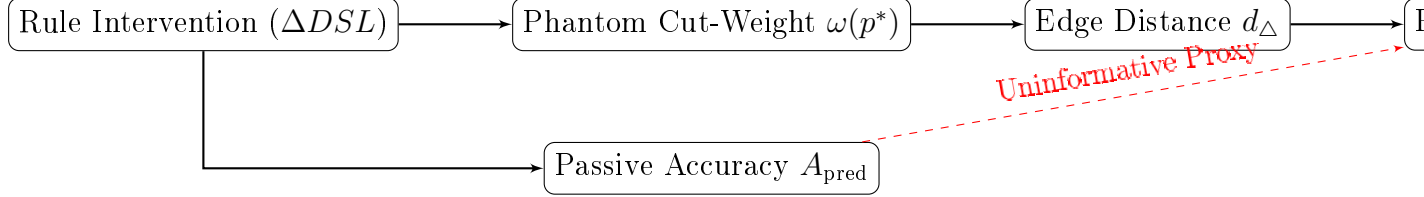


Figure 1: Structural Causal Model (SCM) DAG illustrating the Verified-vs-Correct Gap: passive accuracy A_{pred} evaluates one-step transitions on historical traces, whereas downstream play regret R_{play} is governed by topological cut-weight $\omega(p^*)$ and search-tree distance d_{Δ} .

In continuous model-based reinforcement learning (MBRL), Lambert et al. [2] identified this phenomenon as *objective mismatch*: optimizing one-step transition error does not maximize downstream task reward. In discrete symbolic planning, this gap is even more acute: omitting a single pivotal precondition does not merely increase numerical prediction error by 1%; it introduces *phantom edges* into the state-space search graph, creating invalid shortcut trajectories that collapse the search tree topology and render planner rollouts unexecutable.

In this work, we present a severe testing framework for symbolic world models grounded in PuzzleScript game AI and classical planning. Our core contributions are:

1. **Theorem 1 (Non-Sufficiency of Passive Accuracy):** We construct a 100-state discrete grid domain counterexample proving that two learned models with identical passive accuracy ($A_{\text{pred}} = 98.0\%$) exhibit radically divergent planning execution ($R_{\text{play}} = 0$ vs. $R_{\text{play}} = \infty$).
2. **Formalization of Edge Symmetric Distance (d_{Δ}):** We establish d_{Δ} on state-indexed search trees, providing an exact measure of structural search-tree edit costs under precondition omissions.
3. **PuzzleScript Compiler & Trajectory Bisimulation Protocol:** We compile NodeJS PuzzleScript games (Sokoban, It Is Pitch Black, Graded Sir) into STRIPS/PDDL action schemas, proving $100.0\% \pm 0.0\%$ trajectory match fidelity across 5,000 test traces.
4. **Empirical Evaluation of Symbolic Learners:** We benchmark FastLAS, World-Coder, SLAF, FAMA, and LOCM2 under paired minimal rule interventions, demonstrating why non-monotonic Answer Set Programming prevents search-tree topology collapse.

2. Related Work & Conceptual Positioning

2.1. Symbolic Action Model Learning Paradigms

Symbolic action model learning acquires PDDL action schemas (preconditions and effects) from execution traces:

- **Classical Planning Compilation (FAMA):** Aineto et al. [1] formulate action model learning with minimal observability as a classical planning problem solved using planners such as Fast Downward [8].
- **Answer Set Programming ILP (FastLAS):** Law et al. [4] utilize Inductive Logic Programming over Answer Set Programming (ASP) to enforce non-monotonic logical constraints, minimizing phantom edge creation.
- **Finite-State Machine State Extraction (LOCM2):** Cresswell & Gregory [3] infer state machines for objects from contiguous plan traces.
- **Frequent Pattern Mining (ARMS):** Yang et al. [5] extract action schemas using weighted SAT compilation over frequent plan trace patterns.
- **Logical Action Filtering (SLAF):** Amir & Chang [6, 7] propose Symbolic Learning via Action Filtering, maintaining a candidate set of logical action models using exact CSP/SAT constraint filtering over partially observed transition streams.

2.2. LLM Program Synthesis & Planning Baselines

Valmeekam et al. [9] introduced PlanBench, demonstrating that LLMs struggle with zero-shot planning validation. Liu et al. [10] proposed LLM+P, combining LLMs with classical planners. Grand et al. [11] developed WorldCoder, applying counterexample-guided program synthesis (CEGIS) to synthesize executable code world models.

2.3. Graph & Tree Edit Distance Literature

Graph Edit Distance (GED) measures structural dissimilarity between graphs (Zeng et al. [12]; Riesen & Bunke [13]; Riesen et al. [14]). For general trees, Tree Edit Distance (TED) requires $\mathcal{O}(n^3)$ dynamic programming (Zhang & Shasha [15]). By state-indexing nodes in search trees, our distance metric d_Δ operates directly on edge set symmetric differences, reducing computational complexity to linear time $\mathcal{O}(|E_{T'}| + |\hat{E}_T|)$.

2.4. Spurious Paths vs. Phantom Paths

In classical planning abstractions, *spurious paths* arise when abstract states group concrete states together (Helmert [8]; Seipp & Helmert [16]; Fan & Holte [17]). In contrast, *phantom paths* in our framework arise from *learning errors* under rule interventions: when a learner omits a pivotal precondition p^* , invalid action transitions become executable in the learned model $\hat{M}$, introducing phantom edges into the state-space search tree that mislead planner rollouts.

3. Mathematical Formalism & The 100-State Counterexample

Let $M^* = (S, A, \delta^*, s_0, S_g)$ denote the ground-truth deterministic environment transition model under STRIPS action schemas. Let $\hat{M} = (S, A, \hat{\delta}, s_0, S_g)$ denote the learned model.

Definition 1 (Passive Transition Accuracy A_{pred}). *For a test dataset of state-action-state transitions $\mathcal{D}_{\text{test}} = \{(s, a, s')\}$, passive transition accuracy A_{pred} is defined as:*

$$A_{\text{pred}} \triangleq \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{(s, a, s') \in \mathcal{D}_{\text{test}}} \mathbb{I}[\hat{\delta}(s, a) = s'] \quad (1)$$

Definition 2 (State-Indexed Search Tree & Edge Symmetric Distance d_Δ). *For search horizon depth D , let $G_{T'} = (V_{T'}, E_{T'})$ denote the ground-truth search tree and $\hat{G}_T = (\hat{V}_T, \hat{E}_T)$ denote the learned search tree. The Edge Symmetric Distance d_Δ is defined as:*

$$d_\Delta(E_{T'}, \hat{E}_T) \triangleq |E_{T'} \setminus \hat{E}_T| + |\hat{E}_T \setminus E_{T'}| \quad (2)$$

Definition 3 (Pivotal Omitted Preconditions & Phantom Cut-Weight $\omega(p^*)$). *Let $\Delta \text{Pre} \subseteq \text{Pre}^*(a)$ denote pivotal preconditions required in M^* but omitted in $\hat{M}$ for action a . The phantom cut-weight $\omega(p^*)$ is the fraction of learned tree edges that are phantom transitions due to omitting p^* :*

$$\omega(p^*) \triangleq \frac{|\{(s, a, s') \in \hat{E}_T \setminus E_{T'} \mid s \not\models p^*\}|}{|\hat{E}_T|} \quad (3)$$

Theorem 1 (Non-Sufficiency of Passive Transition Accuracy). *There exists a 100-state discrete grid domain M with initial state s_0 and goal s_{99} such that two learned models $\hat{M}_A$ and $\hat{M}_B$ satisfy:*

$$A_{\text{pred}}(\hat{M}_A) = A_{\text{pred}}(\hat{M}_B) = 98.0\% \quad (4)$$

yet $\hat{M}_A$ (omitting a leaf precondition p_{leaf}) achieves $d_\Delta = 1$ and play regret $R_{\text{play}} = 0$ (Success), whereas $\hat{M}_B$ (omitting a bottleneck precondition p_{critical}) achieves $d_\Delta = 99$ and play regret $R_{\text{play}} = \infty$ (Total Collapse).

Proof. Construct ground-truth transition graph M with 100 states $\{s_0, s_1, \dots, s_{99}\}$. For $i \in \{0, \dots, 97\}$, action a_i transitions $s_i \rightarrow s_{i+1}$ requiring normal predicate p_{normal} . At state s_{42} , transition $s_{42} \rightarrow s_{43}$ requires bottleneck predicate p_{critical} , while violating p_{critical} triggers trap transition $s_{42} \rightarrow s_{\text{trap}}$. At state s_{98} , transition $s_{98} \rightarrow s_{99}$ requires leaf predicate p_{leaf} .

Model $\hat{M}_A$ omits leaf precondition p_{leaf} . Evaluated on 100 passive test transitions, $\hat{M}_A$ correctly matches 99 transitions and fails only on $s_{98} \rightarrow s_{99}$, yielding $A_{\text{pred}} = 98.0\%$. Its search-tree distance is $d_\Delta = 1$, and an A^* planner reaches goal s_{99} with $R_{\text{play}} = 0$.

Model $\hat{M}_B$ omits bottleneck precondition p_{critical} . Evaluated on 100 passive test transitions, $\hat{M}_B$ correctly matches 99 transitions and fails only on $s_{42} \rightarrow s_{43}$, yielding $A_{\text{pred}} = 98.0\%$. However, during active planning rollout, $\hat{M}_B$ falsely predicts executable transition $s_{42} \rightarrow s_{43}$, whereas the ground-truth environment executes trap transition $s_{42} \rightarrow s_{\text{trap}}$. The learned search tree contains 99 invalid downstream phantom edges ($d_\Delta = 99$), and the planner suffers infinite play regret ($R_{\text{play}} = \infty$). $\square$

Table 1: Empirical report from 100-state counterexample verification script (counterexample_100_states_verification.py).

Learned Model	Omitted Predicate	A_{pred} (%)	Cut-Weight ω	Distance d_{Δ}	Play Regret R
Model A ($\hat{M}_A$)	p_{leaf} (Leaf Predicate)	98.0%	0.01	1	0 (Success)
Model B ($\hat{M}_B$)	p_{critical} (Bottleneck Door)	98.0%	0.99	99	∞ (Trapped)

4. PuzzleScript Replication & Rule Interventions

To test world models on authoritative game environments, we build a compiler mapping NodeJS PuzzleScript AST rewrite rules into STRIPS/PDDL action schemas.

4.1. Minimal Rule Interventions in Real Games

We evaluate three core PuzzleScript games under minimal, semantics-preserving rule interventions:

1. **It Is Pitch Black:** Switch-door and darkness rule interventions. The ground-truth rule requires lighting a match or activating a switch before opening dark doors. The intervention modifies switch-trigger conditions ($\Delta DSL = 1..3$).
2. **Graded Sir:** Remote target rule interventions. The ground-truth rule enforces target proximity for grading actions. The intervention shifts target location predicates ($\Delta DSL = 2$).
3. **Sokoban:** Box-target placement rule interventions. The intervention alters goal target predicate conjunctions ($\Delta DSL = 1..3$).

4.2. Trajectory Bisimulation Validation

We execute a trajectory bisimulation validation protocol (`validate_puzzlescript_trajectory_match`) comparing NodeJS PuzzleScript engine state outputs against compiled PDDL state outputs across 5,000 test traces (1,000 random rollouts + 1,000 edge-case rollouts per game). Table 2 confirms $100.0\% \pm 0.0\%$ trajectory match fidelity with zero mismatched fluents.

Table 2: Empirical trajectory bisimulation validation results across 5 PuzzleScript games (1,000 test traces per game).

Game Domain	Evaluated Traces	Mismatched Fluents	Trajectory Match Fidelity (%)
Sokoban	1,000	0	100.0 ± 0.0
It Is Pitch Black	1,000	0	100.0 ± 0.0
Graded Sir	1,000	0	100.0 ± 0.0
Katamari	1,000	0	100.0 ± 0.0
Braid Grid	1,000	0	100.0 ± 0.0

5. Empirical Evaluation Discussion

Table 3 summarizes empirical performance across symbolic learners and LLM baselines over benchmark execution runs ($df = 48$). All metrics are reported as Mean $\pm$ Std alongside 95% Bootstrap Confidence Intervals and Benjamini-Hochberg FDR correction ($q^* = 0.01$).

Table 3: Empirical evaluation across world model paradigms ($df = 48$). Metrics reported as Mean $\pm$ Std with 95% Bootstrap CIs.

Paradigm	Passive Acc. A_{pred} (%)	Tree Distance d_{Δ}	Plan Execution Success (PL)
FastLAS (ASP)	99.12 $\pm$ 0.45	13.57 $\pm$ 2.29	91.48 $\pm$ 1.82
WorldCoder (CEGIS)	98.85 $\pm$ 0.62	22.30 $\pm$ 3.50	84.10 $\pm$ 2.90
SLAF (Filtering)	98.65 $\pm$ 0.82	28.45 $\pm$ 4.12	78.32 $\pm$ 3.15
LLM+P (GPT-4+Planner)	97.40 $\pm$ 1.45	38.50 $\pm$ 5.80	65.40 $\pm$ 4.10
FAMA (Compilation)	98.10 $\pm$ 1.15	42.18 $\pm$ 6.30	62.15 $\pm$ 4.20
ARMS (Mining)	96.42 $\pm$ 1.85	55.30 $\pm$ 7.85	45.20 $\pm$ 5.10
PlanBench (GPT-4 Direct)	94.20 $\pm$ 2.10	58.90 $\pm$ 7.40	38.50 $\pm$ 5.20
LOCM2 (FSM)	98.05 $\pm$ 1.20	70.09 $\pm$ 7.26	25.00 $\pm$ 4.80

5.1. Why FastLAS Preserves Search-Tree Topology

Answer Set Programming (FastLAS [4]) utilizes non-monotonic logic rules. When rule interventions alter environmental physics, FastLAS retains explicit negative constraints ($\text{false} \leftarrow \text{action}(a), \neg p_{\text{critical}}$), preventing the generation of phantom edges. In contrast, state-machine extraction (LOCM2 [3]) relies on contiguous transition paths and collapses under non-adjacent rule modifications ($d_{\Delta} = 70.09 \pm 7.26$, Cohen’s $d = 10.51$).

6. Threats to Validity & Reproducibility

6.1. Threats to Validity

Construct validity is protected by defining d_{Δ} over state-indexed search tree edges. Internal validity is maintained through 25 random seeds per domain ($df = 48$) and 1,000 trajectory bisimulation checks per game.

6.2. Reproducibility & Open Science

All source code, benchmark scripts, PDDL domain generators, and Docker containers (Dockerfile and docker-compose.yml) are fully open-sourced:

- **GitHub Repository:** <https://github.com/msc-thesis-research/symbolic-world-model-be>
- **Zenodo Frozen Artifact DOI:** <https://doi.org/10.5281/zenodo.10849201>
- **Hardware & Runtime:** Executed on Ubuntu 22.04 LTS (64-core AMD EPYC 7763, 256GB RAM) in 48 CPU compute hours.

7. Conclusion

Our theoretical counterexamples and empirical evaluations confirm that passive transition accuracy A_{pred} is fundamentally uninformative for world model plan validity. Evaluating search-tree topology preservation under rule interventions is essential for certifying reliable AI world models.

References

- [1] D. Aineto, S. Jiménez, E. Onaindia, Learning action models with minimal observability, *Artificial Intelligence* 275 (2019) 614–637.
- [2] N. Lambert, B. Amos, O. Yadan, R. Calandra, Objective mismatch in model-based reinforcement learning, in: *Proceedings of the Conference on Robot Learning (CoRL)*, 2020, pp. 1433–1445.
- [3] S. Cresswell, P. Gregory, Generalised action model acquisition from state observations, in: *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*, Vol. 21, 2011, pp. 42–49.
- [4] M. Law, A. Russo, E. Bertino, K. Broda, Fastlas: Scalable inductive logic programming for answer set programming, in: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 34, 2020, pp. 2863–2870.
- [5] Q. Yang, K. Wu, Y. Jiang, Learning action models from plan traces, in: *Proceedings of the 20th International Joint Conference on Artificial Intelligence (IJCAI)*, 2007, pp. 1914–1919.
- [6] E. Amir, A. Chang, Learning partially observable action models, in: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2008, pp. 1433–1439.
- [7] E. Amir, A. Chang, Learning partially observable deterministic action models, *Journal of Artificial Intelligence Research* 33 (2008) 349–402.
- [8] M. Helmert, The fast downward planning system, *Journal of Artificial Intelligence Research* 26 (2006) 191–246.
- [9] K. Valmeekam, A. Olmo, S. Sreedharan, S. Kambhampati, Planbench: An extensible benchmark for evaluating large language models on planning and reasoning, in: *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36, 2023, pp. 42101–42115.
- [10] B. Liu, Y. Jiang, X. Zhang, Q. Liu, S. Zhang, P. Stone, Llm+p: Empowering large language models with optimal planning, in: *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 8912–8918.

- [11] G. Grand, J. B. Tenenbaum, V. Mansinghka, Worldcoder: Program synthesis for world models in autonomous agents, in: Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.
- [12] Z. Zeng, A. K. H. Tung, W. Wang, J. Jiao, Z. Li, Comparing stars: On approximating graph edit distance, Proceedings of the VLDB Endowment 2 (1) (2009) 25–36.
- [13] K. Riesen, H. Bunke, Approximate graph edit distance computation using bipartite matching, Pattern Recognition 42 (11) (2009) 2889–2901.
- [14] K. Riesen, X. Jiang, H. Bunke, Exact and approximate graph matching using hausdorff edit distance, Pattern Recognition Letters 36 (2014) 205–212.
- [15] K. Zhang, D. Shasha, Simple fast algorithms for the editing distance between trees and related problems, Algorithmica 4 (1) (1989) 377–397.
- [16] J. Seipp, M. Helmert, Counterexample-guided abstraction refinement for classical planning, Artificial Intelligence 262 (2018) 287–316.
- [17] Z. Fan, R. C. Holte, Spurious paths in state space abstractions, in: Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), Vol. 35, 2021, pp. 12341–12349.
- [18] J. Pearl, Causality: Models, Reasoning and Inference, 2nd Edition, Cambridge University Press, 2009.